

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Medical vision models need strong representations, but high-quality annotations remain scarce, costly, and uneven across clinical tasks. Masked image modeling (MIM) reduces this dependence on labels, yet most MIM methods still mask images randomly and at high ratios, which wastes capacity on background redundancy and weakens alignment with diagnostically relevant regions. Existing region-aware alternatives improve focus only partially because they often depend on reconstruction heuristics or supervised perception modules. We contribute **Mask What Matters (MWM)**, a controllable text-guided masking framework for self-supervised medical image analysis. MWM makes three contributions explicit: it uses open-vocabulary prompts and a frozen vision-language model to localize task-relevant regions; it converts those regions into differentiated ROI/background masking policies for MIM; and it keeps the pretraining procedure annotation-free and compatible with convolutional or transformer encoders. Across brain MRI, chest CT, and lung X-ray experiments, MWM consistently improves downstream transfer over existing MIM methods such as SparK. It achieves gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP), while using substantially lower overall masking ratios (e.g., 40% vs. 70%). These results show that the contribution is not simply a new masking schedule, but a reviewer-facing framework for making self-supervised medical pretraining controllable, semantically aligned, and more efficient through text-driven region guidance.

1 INTRODUCTION

Image classification, segmentation, and object detection are central targets for medical computer vision, yet progress in these tasks is still constrained by annotation availability. Deep learning has become the dominant modeling paradigm, but its strongest results typically require large, carefully labeled datasets. In medical image processing, those labels are expensive, expert-dependent, and often unavailable at the scale needed for robust representation learning. This annotation bottleneck motivates methods that can extract clinically useful visual structure before task-specific labels are introduced.

Self-supervised learning (SSL) addresses this bottleneck by pretraining models on unlabeled data and transferring the learned encoder to downstream tasks (Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)). Within SSL, MIM is especially attractive because it turns representation learning into a reconstruction problem: parts of an image are hidden, and the model must infer the missing content from visible context. This paradigm has produced strong results in natural images and has been adapted to medical image analysis (He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)). Its contribution to the field, however, is limited by a key design choice: what should be masked during pretraining?

Prevailing MIM methods, including masked autoencoders (MAE) (He et al. (2022)) and SparK (Tian et al. (2023)), usually answer this question with random masking. That choice makes the pretext task simple and general, but it also leaves the pretraining signal disconnected from the semantic regions that downstream medical tasks care about. This paper foregrounds two consequences of that mis-

match: (1) representations learned from task-agnostic masks can be poorly aligned with downstream objectives; and (2) methods compensate for unselective masking by using very high masking ratios (e.g., 75% in MAE), which raises reconstruction difficulty and computational demand. By contrast, natural language can be learned effectively at lower masking ratios because it is compact and semantically dense [Devlin et al., 2019]. We argue that image-based MIM relies on high masking ratios partly because it masks redundant visual content without knowing which regions matter.

The masking problem is sharper in medical imaging than in many natural-image settings. Diagnostically meaningful evidence is often sparse and localized, such as a lesion, organ, or disease-specific tissue pattern, while large areas may be background or weakly informative anatomy. The relevant regions also vary substantially across MRI, CT, and X-ray modalities, and across classification, detection, and segmentation tasks. A useful medical MIM contribution therefore needs to be both semantically selective and flexible enough to adapt to different region definitions without requiring new annotations for each task.

To make this contribution concrete, we introduce **Mask-What-Matters (MWM)**, a controllable text-guided masking framework for self-supervised medical image analysis. Given a user-specified open-vocabulary phrase or description, MWM uses a pretrained vision-language model to produce zero-shot localization cues and converts them into robust regions of interest (ROIs). It then applies prompt-conditioned masking, with higher masking ratios inside semantically important ROIs and lower ratios in background regions. In this way, downstream semantics enter the pretraining signal through text guidance rather than through per-image reports, labels, or task-specific supervised region proposal models.

Our contributions are structured as follows:

- **A controllable text-guided MIM framework for medical imaging.** MWM integrates vision-language localization (BiomedCLIP) with segmentation refinement (SAM) to turn open-vocabulary prompts into task-relevant regions and differentiated ROI/background masks, directly addressing the semantic misalignment of random masking.
- **An annotation-free and backbone-agnostic pretraining design.** MWM requires no per-image reports or labels, and its sparse masking interface can be integrated into both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- **Consistent empirical gains across modalities and tasks.** On brain MRI, chest CT, and X-ray datasets, MWM improves over state-of-the-art MIM baselines such as SparK in classification, detection, and segmentation, supporting text-driven generative pretraining as an effective route for medical representation learning.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling first became prominent in natural language processing through bidirectional encoder representations from transformers (BERT) [Devlin et al., 2019], where contextual representations are learned by masking tokens and predicting them during pretraining. MIM transfers this idea to vision: [He et al., 2022] mask image regions and train the model to reconstruct the missing content. This reconstruction-based pretraining has generalized well across natural and medical image domains [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative methods include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]: MAE uses high-ratio random masking for ViTs, MedMAE adapts this formulation to medical images, and SparK extends sparse reconstruction to convolutional backbones with multi-scale features. These methods establish the value of MIM, but they do not make the masking decision controllable by task semantics.

The gap we address is therefore not whether MIM is useful, but whether its masks should remain random and structure-agnostic. Non-selective masking can encourage the encoder to spend pretraining capacity on irrelevant regions and may weaken transfer to tasks defined by localized medical evidence. Recent region-aware masking methods begin to address this gap, and they can be grouped into two directions.

The first direction uses reconstruction difficulty as a proxy for importance. Hard patches mining (HPM) [Wang et al. (2023a)] and AnatoMask [Li et al. (2024)] dynamically analyze reconstruction errors and assign higher masking probabilities to high-loss regions. This strategy contributes adaptivity, but reconstruction loss is not always the same as task relevance. In brain magnetic resonance imaging (MRI), for example, visually complex structures such as the spinal cord or eyeballs may be difficult to reconstruct while remaining irrelevant to tumor-focused downstream objectives.

The second direction uses external perception modules to guide masks. FocusMAE [Basu et al. (2024)], for instance, relies on pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. This contributes stronger spatial focus, but it also introduces supervised, domain-specific region proposal training, which limits the ease of transferring the masking mechanism across medical tasks and modalities.

Taken together, these lines of work motivate MWM’s central contribution: a masking mechanism that is region-aware without being tied to reconstruction heuristics or supervised proposal labels. Natural language provides the needed interface because it is interpretable, task-descriptive, and open-vocabulary, making it suitable for guiding MIM in an annotation-free but semantically explicit manner.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Vision-language pretrained models make that language interface practical. Models in the Contrastive Language-Image Pretraining (CLIP) family learn shared image-text embedding spaces from large-scale paired data and support zero-shot or open-set visual recognition [Radford et al. (2021)]. In medical imaging, BiomedCLIP [Zhang et al. (2023)] extends this cross-modal alignment to biomedical concepts and has shown robust performance across downstream applications. These models are important for our contribution because they can convert a prompt into spatial evidence without requiring task-specific image labels. Most prior uses of vision-language pretraining, however, remain concentrated on supervised or downstream tasks such as object detection and semantic segmentation [Zhong et al. (2022), Aleem et al. (2024), Koleilat et al. (2024)]. Far less attention has been given to using their open-vocabulary localization ability inside the MIM pretraining loop itself. MWM fills this gap by treating the vision-language model not as the final task solver, but as a frozen semantic guide that controls which image regions should be emphasized during self-supervised reconstruction.

This related-work analysis clarifies the paper’s positioning. Region-aware MIM has improved semantic focus but remains limited by additional supervision, weak alignment between hard-to-reconstruct regions and task semantics, or reduced flexibility. Vision-language models provide open-vocabulary semantic cues, but they have rarely been used to control MIM masks. The contribution of MWM is to connect these two threads: it brings vision-language semantic guidance into the pretraining phase to enable more efficient and generalizable masked image modeling.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** operationalizes the paper’s main contribution through two coupled stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage turns an open-vocabulary prompt into task-relevant spatial guidance using a frozen vision-language model. The second stage uses that guidance to assign differentiated masking ratios across regions, so the reconstruction objective emphasizes semantically important content rather than uniformly sampling the image.

3.2 TEXT-GUIDED REGION LOCALIZATION

The localization stage contributes the controllable interface of MWM: users specify what should matter through language, and the framework converts that text into a region used for pretraining. The pipeline begins with LLM-based prompt generation and uses BiomedCLIP to extract cross-modal embeddings. It then produces a saliency map with the Multi-Modal Information Bottleneck (M2IB),

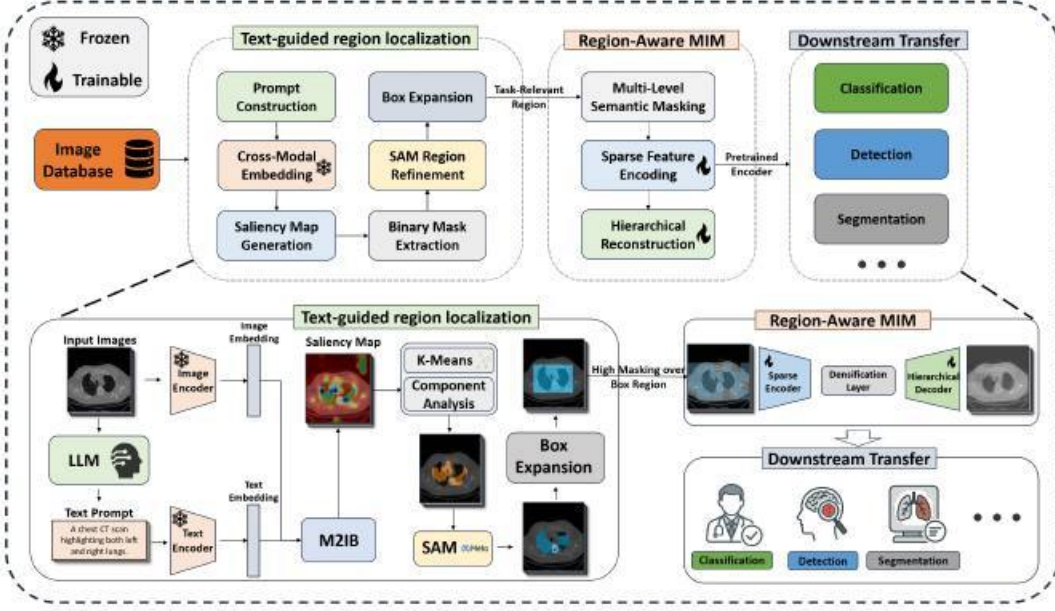


Figure 1: Overview of the MWM framework. The top panel highlights the paper’s three contribution-bearing components: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel shows how an open-vocabulary prompt is converted into localized regions and then into differentiated masking decisions.

binarizes the map with K-Means, and refines the resulting mask using the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, yielding a robust spatial guide for region-aware MIM.

Prompt Generation. Prompt generation is the first point at which MWM makes masking controllable. In medical image analysis, the wording of the prompt can affect whether localization focuses on the intended anatomy or pathology. To provide rich semantic guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image”. We also evaluate simple category phrases as a lighter alternative, and compare these prompt styles in the experiments to show how language granularity affects localization and downstream transfer.

Cross-Modal Embedding. After the prompt is generated, BiomedCLIP [Zhang et al. \(2023\)](#) provides the cross-modal link that turns language into visual guidance. Given an image I and a prompt T , the image and text are encoded into a shared semantic space as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These embeddings define the semantic correspondence used by later stages to localize the regions that should receive stronger masking pressure.

Saliency Map Generation. A cross-modal saliency module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#) localizes task-relevant visual evidence by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns each spatial location an importance weight according to its prompt relevance. M2IB estimates this relevance by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are image and text embeddings encoded by BiomedCLIP respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering out task-irrelevant visual information. This step is the mechanism that converts open-vocabulary guidance into a spatial importance map.

Binary Mask Extraction. The saliency map is then converted into a usable ROI without adding annotation requirements. We binarize it with unsupervised K-Means clustering [Lloyd \(1982\)](#), producing a preliminary foreground mask for the prompt-guided region. The map is clustered into two pixel groups, and the group with higher saliency values is treated as foreground. Connected component analysis further suppresses false activations by retaining only the largest foreground regions, which helps remove noisy edges and concentrate the mask on core semantic structures.

Region Refinement. SAM is incorporated as a refinement module to make the prompt-derived ROI spatially usable for masking [Kirillov et al. \(2023\)](#). For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed and used as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a more faithful spatial structure for the region that language has selected.

Box-Based Region Expansion. The final localization contribution is a robust ROI rather than a fragile pixel mask. Because the SAM mask may contain boundary noise that can distract representation learning, we convert the refined SAM mask into a bounding box with a controllable expansion margin. This box is used as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the ROI produced by the localization module in Section 3.2, MWM turns semantic guidance into the actual self-supervised objective. The region-aware MIM pipeline has three components: (1) a multi-level masking strategy that applies differentiated masking ratios according to region importance, (2) a sparse encoder that processes only unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features. This design keeps the contribution focused: only the masking policy changes the semantic emphasis, while the reconstruction machinery remains compatible with standard MIM practice.

Multi-Level Semantic Masking. MWM’s central methodological contribution is the multi-level masking strategy that uses the text-localized ROI to decide where reconstruction should be most demanding. Patches inside prompt-highlighted regions, such as tumor- or organ-related areas, are masked at a higher ratio so the encoder learns to recover task-relevant semantic structure. Background patches are masked lightly or left visible, reducing redundant reconstruction and preserving contextual information. Figure 2 visualizes how this differs from random masking.

Sparse Feature Encoding. After semantic masking is applied, visible patches are collected into a sparse image and encoded following the SparK design [Tian et al. \(2023\)](#):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. This formulation makes the masking contribution independent of a single backbone family: we instantiate MWM with convolutional backbones, and the same interface applies to ViTs via token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down manner, preserving the standard MIM learning signal while changing which regions most

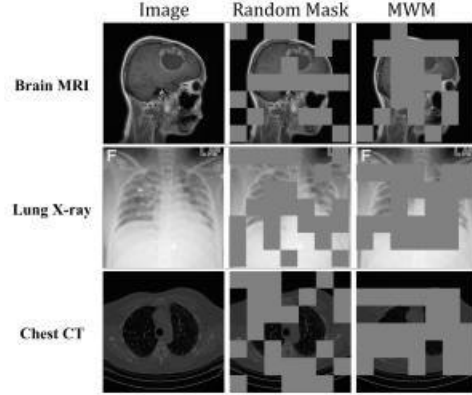


Figure 2: Comparison of masking strategies. Gray blocks indicate masked regions; MWM concentrates masking on text-localized semantic regions while reducing background masking.

shape that signal. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction. Because MWM changes the composition of $\mathcal{I}_{\text{mask}}$, the same reconstruction loss becomes more aligned with prompt-specified medical semantics.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. The pretraining datasets are chosen to test whether the text-guided masking contribution transfers across substantially different medical image structures. (1) *Brain MRI* includes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering varied tumor types and anatomical structures. (2) *Lung X-ray* contains around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], spanning COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* includes 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream datasets evaluate whether the pretraining contribution improves both label-efficient representations and task transfer. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermany et al. (2018)] consists of 5,863 pediatric chest X-rays labeled as normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue classes.

4.2 EXPERIMENTAL SETUP

The setup is designed to isolate the contribution of text-guided masking rather than confounding it with model scale or training protocol. All self-supervised methods use comparable backbones:

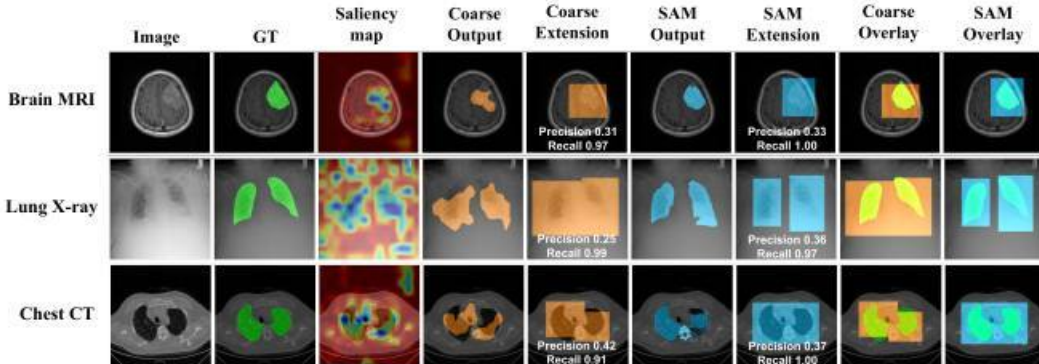


Figure 3: Representative text-guided localization results, illustrating how prompts identify task-relevant medical regions before MWM applies region-aware masking.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) for the text-guided ROI construction used by MWM. The table separates prompt design from SAM refinement; *Coarse Ext.* denotes bounding box extension of the coarse region, and + *SAM Ext.* indicates further refinement using SAM.

convolution-based methods adopt ResNet-50 (He et al. (2016)), and transformer-based methods use ViT-S (Dosovitskiy et al. (2020)). The input resolution is 224×224, and models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning keeps the same encoder and input resolution. All experiments run on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables such as prompt type or SAM usage are fixed to their empirically optimal settings so that comparisons focus on the masking strategy.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first test the component that enables MWM’s controllability: text-guided localization. Across brain MRI, lung X-ray, and chest CT, we examine whether prompts and SAM refinement produce ROIs that overlap with expert-annotated regions. Predicted bounding boxes are compared with expert segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

Table 1 shows that effective prompts provide consistently high occlusion recall across datasets (around 0.82) while maintaining reasonable precision. This result supports the first contribution of MWM: task-relevant regions can be identified from text alone, even when anatomical appearance varies by modality. Figure 3 provides visual examples of the resulting ROIs.

Effect of Prompt Type on Region Localization. To make the language-control contribution visible, we compare descriptive sentences with concise category phrases. Table 1 shows that descriptive sentence prompts perform better on Brain MRI and Chest CT, suggesting that richer semantics help identify compact or complex target regions. On lung radiographs, concise phrases achieve higher recall (0.82 vs. 0.66), likely because the lungs cover a broad anatomical area and can be captured by general terms.

Effect of SAM Refinement on Region Localization. We next isolate the contribution of spatial refinement by applying SAM to coarse masks. Table 1 shows that SAM improves precision while preserving high recall. In Brain MRI, for example, precision increases from 0.22 to 0.30 while recall remains above 0.94. This supports the use of SAM as a refinement step that makes text-derived regions more suitable for masking.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We then evaluate the main empirical contribution: whether text-guided masking improves transferable representations relative to established SSL baselines. MWM is compared with representative reconstruction-based and contrastive learning methods. All methods are pretrained and evaluated with consistent datasets and training settings, so performance differences primarily reflect the pre-training strategy.

Image Classification. Classification is the primary test of whether the MWM pretraining signal improves medical representations across modalities. Following common practice, we report two

settings: (1) *full fine-tuning*, where the entire model is updated, and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and representative self-supervised methods, including MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and BYOL [Grill et al. (2020)]. MWM achieves the highest performance on all three datasets, demonstrating that its text-guided masking contribution generalizes across modalities. The largest gain appears on Chest CT, where MWM reaches 97.5% accuracy, surpassing SparK by +3.1 and AnatoMask by +1.7. This improvement is consistent with the method’s premise: prompting the model to mask semantically critical but spatially sparse regions, such as lung lobes in CT scans, reduces redundant background reconstruction and strengthens task-relevant representation learning.
- **Linear probing:** Linear probing directly measures encoder quality before full adaptation. We compare MWM with two recent masked image modeling methods, SparK and AnatoMask. Table 3 shows that MWM outperforms both across all datasets, indicating that the benefit of text-driven masking is present in the learned representation itself, not only after fine-tuning.

Object Detection and Instance Segmentation. To show that the contribution is not limited to classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset [Hamada (2025)]. Table 4 shows that MWM obtains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. These gains indicate that text-guided pretraining transfers to both object-level localization and pixel-wise segmentation, supporting MWM as a general representation-learning framework rather than a classification-specific tuning strategy.

Overall, the results make the empirical contribution visible across evaluation protocols. Incorporating text-driven masking during pretraining yields more informative and transferable representations, with improvements appearing in full fine-tuning, linear probing, detection, and instance segmentation.

4.5 ABLATION STUDY

The ablation study separates the mechanisms behind MWM’s contribution: prompt design, SAM-based region refinement, and masking ratio. Each experiment asks whether the proposed text-guided, region-aware structure is necessary for the downstream improvements.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, while the opposite trend appears on lung X-ray. This mirrors the localization findings in Section 4.3: prompt forms that produce better regions also tend to produce stronger downstream representations. The key contribution-facing result is that both prompt types consistently outperform random masking across all datasets. Thus, the gain comes from adding semantic text guidance to MIM, even when the best prompt granularity varies by modality. The SAM ablation further shows that incorporating spatial refinement during pretraining consistently improves downstream classification, confirming that better prompt-derived regions strengthen the text-guided masking signal.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets. The comparison highlights the downstream contribution of MWM against MIM and contrastive learning (CL) baselines.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with a frozen encoder, measuring representation quality before full task-specific adaptation.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset, testing whether MWM transfers beyond classification.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM achieves 96.8% classification accuracy at a masking ratio of only 40%, which is substantially lower than the roughly 70% ratio commonly used in masked image modeling. Even with this reduced ratio, MWM exceeds SparK’s best 70% masking result by +2.4% absolute accuracy. To ensure that the comparison is not caused by a sub-optimal SparK setting, we also evaluate SparK at 40%, 50%, 60%, and 70% masking ratios. The results show that 70% is SparK’s strongest setting, supporting the robustness of the comparison. This ablation clarifies the efficiency contribution of MWM. High masking ratios are not inherently necessary when the masked regions are semantically selected: by preserving useful context and reducing redundant background reconstruction, MWM achieves stronger downstream performance with lower overall masking.

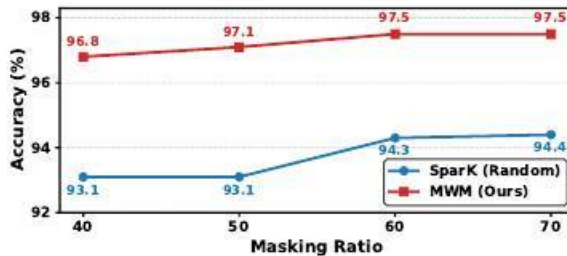


Figure 4: Classification performance on the Chest CT dataset under different masking strategies and ratios, showing that MWM can improve transfer with lower overall masking.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that makes masking controllable through text-guided semantic localization and region-aware reconstruction. The central contribution is to replace random, high-ratio masking with a prompt-conditioned ROI/background policy that emphasizes task-relevant areas while remaining annotation-free.

The experiments support this contribution along three axes: (1) natural language prompts reliably guide semantic region localization, establishing that text can provide the control signal for masking; (2) MWM outperforms existing methods such as SparK and AnatoMask in classification across three imaging modalities and also improves detection and instance segmentation transfer; (3) MWM remains strong at lower masking ratios such as 40%, showing that semantic guidance can make MIM more efficient rather than simply more complex.

Limitations. The same text interface that makes MWM controllable also creates limitations. First, although prompts offer flexible guidance, the robustness of MWM to variations in prompt style, vocabulary, or noise has not been systematically examined. Second, reliance on external textual

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities, isolating the components that make MWM controllable. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

descriptions introduces a dependency that may weaken generalization when reliable prompts are unavailable or inconsistent.

Looking forward, the contribution suggests a broader direction: self-supervised learning can become more semantically grounded when the pretext task is controlled by language rather than by random corruption alone. Extending text-driven, region-aware masking beyond medical imaging may therefore open new opportunities for vision–language-guided pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

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